نتایج ارزیابی برای دیتاست CIFAR10:

Batch Size = 150

Epoch = 300

Time of training = about 8 hours

Training from Scratch (Without using pretrain weights)

Number of Test data = 10000

Number of Training data= 50000

مقدار رم GPU 4060 درگیر در حالت SNNحدودا 16 گیگ بوده ولی با همین تنظیمات در حالت ANN حجم اشغالی حدودا 23 گیگ بود.

Test: [0/45] eta: 0:09:37 loss: 0.2137 (0.2137) acc1: 95.5556 (95.5556) acc5: 99.5556 (99.5556) time: 12.8435 data: 11.7839 max mem: 3920

Test: [10/45] eta: 0:00:48 loss: 0.2137 (0.2350) acc1: 95.5556 (95.2323) acc5: 99.5556 (99.6768) time: 1.3763 data: 1.0714 max mem: 3920

Test: [20/45] eta: 0:00:20 loss: 0.2401 (0.2415) acc1: 94.6667 (94.9630) acc5: 99.5556 (99.7249) time: 0.2197 data: 0.0001 max mem: 3920

Test: [30/45] eta: 0:00:09 loss: 0.2405 (0.2407) acc1: 95.1111 (95.2545) acc5: 99.5556 (99.6703) time: 0.2074 data: 0.0001 max mem: 3920

Test: [40/45] eta: 0:00:02 loss: 0.2400 (0.2406) acc1: 95.5556 (95.3171) acc5: 99.5556 (99.6748) time: 0.1954 data: 0.0001 max mem: 3920

Test: [44/45] eta: 0:00:00 loss: 0.2400 (0.2414) acc1: 95.5556 (95.2900) acc5: 99.5556 (99.6800) time: 0.1899 data: 0.0001 max mem: 3920

Test: Total time: 0:00:22 (0.4941 s / it)

* Acc@1 95.290 Acc@5 99.680 loss 0.241

Accuracy of the network on 10000 test images: 95.29000%

نتایج بررسی **Sparsity** :

Batch [1/67] Accuracy: 96.00% | Avg Layer Sparsity: 33.20%

Batch [10/67] Accuracy: 95.00% | Avg Layer Sparsity: 33.22%

Batch [20/67] Accuracy: 95.03% | Avg Layer Sparsity: 33.33%

Batch [30/67] Accuracy: 94.96% | Avg Layer Sparsity: 32.98%

Batch [40/67] Accuracy: 95.18% | Avg Layer Sparsity: 33.33%

Batch [50/67] Accuracy: 95.29% | Avg Layer Sparsity: 33.18%

Batch [60/67] Accuracy: 95.32% | Avg Layer Sparsity: 33.31%

==========

FINAL EVALUATION RESULTS For SPARSITY

==========

Overall Accuracy: 95.29%

Overall Average Sparsity: 0.33%

Total samples evaluated: 7500

Per-Layer Average Sparsities:

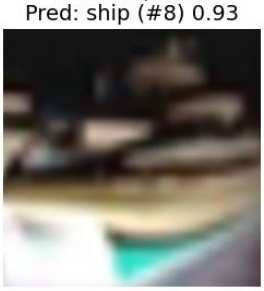
downsample_layers: 0.00%

head: 99.64%

stages: 0.00%

True: cat (#3)
Pred: cat (#3) 0.92

True: ship (#8)
Pred: ship (#8) 0.93



True: ship (#8)
Pred: ship (#8) 0.52

True: airplane (#0)
Pred: airplane (#0) 0.92

True: frog (#6)
Pred: frog (#6) 0.93

True: frog (#6)
Pred: frog (#6) 0.92

True: automobile (#1)
Pred: automobile (#1) 0.92

True: frog (#6)
Pred: frog (#6) 0.87

True: cat (#3)
Pred: cat (#3) 0.92

True: automobile (#1)
Pred: automobile (#1) 0.88

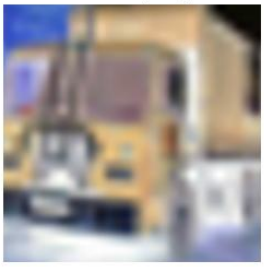
True: airplane (#0)
Pred: airplane (#0) 0.83

True: truck (#9)
Pred: truck (#9) 0.92

True: dog (#5)
Pred: dog (#5) 0.84

True: horse (#7)
Pred: horse (#7) 0.95

True: truck (#9)
Pred: truck (#9) 0.92



True: ship (#8)
Pred: ship (#8) 0.90

```
==========================================================================================
Layer (type:depth-idx)                   Output Shape              Param #
==========================================================================================
ConvNeXtSpiking                          [1, 10]                   --
├─ModuleList: 1-7                        --                        (recursive)
│    └─Sequential: 2-1                   [1, 96, 8, 8]             --
│    │    └─Conv2d: 3-1                  [1, 96, 8, 8]             4,704
├─ModuleList: 1-8                        --                        (recursive)
│    └─Sequential: 2-2                   [1, 96, 8, 8]             --
│    │    └─SpikingBlock: 3-2            [1, 96, 8, 8]             79,104
│    │    └─SpikingBlock: 3-3            [1, 96, 8, 8]             79,104
│    │    └─SpikingBlock: 3-4            [1, 96, 8, 8]             79,104
├─ModuleList: 1-7                        --                        (recursive)
│    └─Sequential: 2-3                   [1, 192, 4, 4]            --
│    │    └─Conv2d: 3-5                  [1, 192, 4, 4]            73,920
├─ModuleList: 1-8                        --                        (recursive)
│    └─Sequential: 2-4                   [1, 192, 4, 4]            --
│    │    └─SpikingBlock: 3-6            [1, 192, 4, 4]            305,664
│    │    └─SpikingBlock: 3-7            [1, 192, 4, 4]            305,664
│    │    └─SpikingBlock: 3-8            [1, 192, 4, 4]            305,664
├─ModuleList: 1-7                        --                        (recursive)
│    └─Sequential: 2-5                   [1, 384, 2, 2]            --
│    │    └─Conv2d: 3-9                  [1, 384, 2, 2]            295,296
├─ModuleList: 1-8                        --                        (recursive)
│    └─Sequential: 2-6                   [1, 384, 2, 2]            --
│    │    └─SpikingBlock: 3-10           [1, 384, 2, 2]            1,201,152
│    │    └─SpikingBlock: 3-11           [1, 384, 2, 2]            1,201,152
│    │    └─SpikingBlock: 3-12           [1, 384, 2, 2]            1,201,152
│    │    └─SpikingBlock: 3-13           [1, 384, 2, 2]            1,201,152
│    │    └─SpikingBlock: 3-14           [1, 384, 2, 2]            1,201,152
│    │    └─SpikingBlock: 3-15           [1, 384, 2, 2]            1,201,152
│    │    └─SpikingBlock: 3-16           [1, 384, 2, 2]            1,201,152
│    │    └─SpikingBlock: 3-17           [1, 384, 2, 2]            1,201,152
│    │    └─SpikingBlock: 3-18           [1, 384, 2, 2]            1,201,152
├─ModuleList: 1-7                        --                        (recursive)
│    └─Sequential: 2-7                   [1, 768, 1, 1]            --
│    │    └─Conv2d: 3-19                 [1, 768, 1, 1]            1,180,416
├─ModuleList: 1-8                        --                        (recursive)
│    └─Sequential: 2-8                   [1, 768, 1, 1]            --
│    │    └─SpikingBlock: 3-20           [1, 768, 1, 1]            4,761,600
│    │    └─SpikingBlock: 3-21           [1, 768, 1, 1]            4,761,600
│    │    └─SpikingBlock: 3-22           [1, 768, 1, 1]            4,761,600
├─Linear: 1-9                            [1, 10]                   7,690
Trainable params: 27,811,498
Non-trainable params: 0
Total mult-adds (M): 6.04
==========================================================================================
Input size (MB): 0.01
Forward/backward pass size (MB): 0.44
Params size (MB): 7.57
Estimated Total Size (MB): 8.03
==========================================================================================
```

```
Model = ConvNeXtSpiking(
 (downsample_layers): ModuleList(
  (0): Sequential(
   (0): Conv2d(3, 96, kernel_size=(4, 4), stride=(4, 4))
  )
  (1): Sequential(
   (0): Conv2d(96, 192, kernel_size=(2, 2), stride=(2, 2))
  )
  (2): Sequential(
   (0): Conv2d(192, 384, kernel_size=(2, 2), stride=(2, 2))
  )
  (3): Sequential(
   (0): Conv2d(384, 768, kernel_size=(2, 2), stride=(2, 2))
  )
 )
 (stages): ModuleList(
  (0): Sequential(
   (0): SpikingBlock(
    (dwconv): Conv2d(96, 96, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=96)
    (pw1): Linear(in_features=96, out_features=384, bias=True)
    (pw2): Linear(in_features=384, out_features=96, bias=True)
    (drop_path): Identity()
   )
   (1): SpikingBlock(
    (dwconv): Conv2d(96, 96, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=96)
    (pw1): Linear(in_features=96, out_features=384, bias=True)
    (pw2): Linear(in_features=384, out_features=96, bias=True)
    (drop_path): Identity()
   )
   (2): SpikingBlock(
    (dwconv): Conv2d(96, 96, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=96)
```

```
    (pw1): Linear(in_features=96, out_features=384, bias=True)
    (pw2): Linear(in_features=384, out_features=96, bias=True)
    (drop_path): Identity()
   )
  )
  (1): Sequential(
   (0): SpikingBlock(
    (dwconv): Conv2d(192, 192, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=192)
    (pw1): Linear(in_features=192, out_features=768, bias=True)
    (pw2): Linear(in_features=768, out_features=192, bias=True)
    (drop_path): Identity()
   )
   (1): SpikingBlock(
    (dwconv): Conv2d(192, 192, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=192)
    (pw1): Linear(in_features=192, out_features=768, bias=True)
    (pw2): Linear(in_features=768, out_features=192, bias=True)
    (drop_path): Identity()
   )
   (2): SpikingBlock(
    (dwconv): Conv2d(192, 192, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=192)
    (pw1): Linear(in_features=192, out_features=768, bias=True)
    (pw2): Linear(in_features=768, out_features=192, bias=True)
    (drop_path): Identity()
   )
  )
  (2): Sequential(
   (0): SpikingBlock(
    (dwconv): Conv2d(384, 384, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=384)
    (pw1): Linear(in_features=384, out_features=1536, bias=True)
    (pw2): Linear(in_features=1536, out_features=384, bias=True)
    (drop_path): Identity()
```

)

(1): SpikingBlock(

(dwconv): Conv2d(384, 384, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=384)

(pw1): Linear(in_features=384, out_features=1536, bias=True)

(pw2): Linear(in_features=1536, out_features=384, bias=True)

(drop_path): Identity()

)

(2): SpikingBlock(

(dwconv): Conv2d(384, 384, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=384)

(pw1): Linear(in_features=384, out_features=1536, bias=True)

(pw2): Linear(in_features=1536, out_features=384, bias=True)

(drop_path): Identity()

)

(3): SpikingBlock(

(dwconv): Conv2d(384, 384, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=384)

(pw1): Linear(in_features=384, out_features=1536, bias=True)

(pw2): Linear(in_features=1536, out_features=384, bias=True)

(drop_path): Identity()

)

(4): SpikingBlock(

(dwconv): Conv2d(384, 384, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=384)

(pw1): Linear(in_features=384, out_features=1536, bias=True)

(pw2): Linear(in_features=1536, out_features=384, bias=True)

(drop_path): Identity()

)

(5): SpikingBlock(

(dwconv): Conv2d(384, 384, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=384)

(pw1): Linear(in_features=384, out_features=1536, bias=True)

(pw2): Linear(in_features=1536, out_features=384, bias=True)

(drop_path): Identity()

)

```
  (6): SpikingBlock(
    (dwconv): Conv2d(384, 384, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=384)
    (pw1): Linear(in_features=384, out_features=1536, bias=True)
    (pw2): Linear(in_features=1536, out_features=384, bias=True)
    (drop_path): Identity()
  )
  (7): SpikingBlock(
    (dwconv): Conv2d(384, 384, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=384)
    (pw1): Linear(in_features=384, out_features=1536, bias=True)
    (pw2): Linear(in_features=1536, out_features=384, bias=True)
    (drop_path): Identity()
  )
  (8): SpikingBlock(
    (dwconv): Conv2d(384, 384, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=384)
    (pw1): Linear(in_features=384, out_features=1536, bias=True)
    (pw2): Linear(in_features=1536, out_features=384, bias=True)
    (drop_path): Identity()
  )
)
(3): Sequential(
  (0): SpikingBlock(
    (dwconv): Conv2d(768, 768, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=768)
    (pw1): Linear(in_features=768, out_features=3072, bias=True)
    (pw2): Linear(in_features=3072, out_features=768, bias=True)
    (drop_path): Identity()
  )
  (1): SpikingBlock(
    (dwconv): Conv2d(768, 768, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=768)
    (pw1): Linear(in_features=768, out_features=3072, bias=True)
    (pw2): Linear(in_features=3072, out_features=768, bias=True)
    (drop_path): Identity()
```

```
    )
    (2): SpikingBlock(
      (dwconv): Conv2d(768, 768, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=768)
      (pw1): Linear(in_features=768, out_features=3072, bias=True)
      (pw2): Linear(in_features=3072, out_features=768, bias=True)
      (drop_path): Identity()
    )
   )
  )
  (head): Linear(in_features=768, out_features=10, bias=True)
)
number of params: 27811498
LR = 0.00040000
Batch size = 150
Update frequent = 1
Number of training examples = 50000
Number of Epoch: 300
Number of training training per epoch (iteration) = 333
```